



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: **CSC418**

Credit Hours: **3(2,1)**

Lab Hours/Week: **3**

Course Title: **DevOps for Cloud Computing**

Lecture Hours/Week: **2**

Pre-Requisites: **None**

Catalogue Description:

This course introduces the core concepts of implementing DevOps using cloud computing. Topics include: Cloud Computing; Service Models; Multi-Cloud; Deployment Models; Emerging Cloud Trends and Practices; Virtualization & Containerization; Components of a Cloud Infrastructure; Introduction to DevOps and its Roles; Source Control Management; DevOps Utility and Automation; CI/CD; Infrastructure as Code (IAC); SDLC of DevOps; Microservices; Serverless Computing; DevOps Cloud Native; Application Modernization; Deployment to Cloud; Operate and Monitor using DevOps Tools.

Mapping of CLOs and SOs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO's for Theory				
CLO-1	1	Characterize the fundamentals of cloud computing along with its service models.	<i>Understanding</i>	1
CLO-2	2	Enumerate virtualization and containerization techniques used in cloud computing.	<i>Understanding</i>	1
CLO-3	3-5	Compare centralized and distributed types of source control management techniques.	<i>Analyzing</i>	2,4
CLO's for Lab				
CLO-4	4	Apply DevOps pipeline automation techniques for code deployment	<i>Applying</i>	2,4
CLO-5	5	Develop cloud native applications using current DevOps tools.	<i>Creating</i>	3-5, 9

Week Wise Plan

Lecture #	CDF Unit #	Topics Covered	Textbook Section
1.	1	Traditional Development Vs DevOps Culture; Role of Cloud in DevOps.	Verona: Ch1
2.	1	Introduction to Cloud Computing; Cloud Service Models: IaaS, PaaS, SaaS	Sehgal: Ch1
3.	1	Essential Characteristics of Cloud Computing; Emerging Cloud Trends and Practices; Components of a Cloud Infrastructure	Sehgal: Ch1, Ch2
4.	1	Deployment Models; Features of Private and Public Cloud; Hybrid Clouds	Sehgal: Ch4

Quiz 01			
5.	2	Introduction to Virtualization; Role of Virtualization in Cloud; Introduction to Hypervisors; Types of Hypervisors; Creation of a Linux based Virtual Machine	Sehgal: Ch3
6.	2	Linux OS Introduction; Importance of Linux in DevOps; Linux Basic Command Utilities; Package Manager	Verona: Ch2;
7.	2	User Management and System Administration in Linux	Online Resource [1]
8.	2	Introduction to Amazon Web Services; Amazon Elastic Compute Cloud (EC2); Launching a Linux based EC2 Instance; Using SSH client to connect to EC2 Instance.	Online Resource [2]
Assignment 01			
9.	2	Monolithic vs Microservice Architecture; Introduction to Containerization; Containerization vs Virtualization; Docker as Container Engine;	Verona: Ch3; Online Resource [3]
10.	2	Docker Commands; Introduction to <i>dockerfile</i> ; Creating Custom Images using <i>dockerfile</i> ;	Online Resource [3]
11.	2	Introduction to <i>docker-compose</i> ; Setting up Containerized Environment using <i>docker-compose</i>	Online Resource [3]
12.	3	DevOps Concepts; Agile and DevOps; The DevOps Process and Continuous Integration	Verona: Ch1, Ch2
13.	3	Code Stage; Use of Source Code Management Systems (SCM); Distributed vs Centralized SCM	Verona: Ch4
14.	3	Git as a Tool for Source Code Management; Git Commands	Verona: Ch4; Online Resource [4]
15.	3	GitHub; GitHub Action; GitHub Webhooks	Online Resource [4]
Quiz 02, Assignment 02			
16.	4	Introduction to Jenkins; A Continuous Integration / Continuous Delivery (CI/CD) Tool	Verona: Ch5; Online Resource [5]
17.	4	Jenkins Plugins; Setting up a Job in Jenkins having <i>Build</i> Stage over AWS with GitHub Integration	Online Resource [5]
Mid Term Exam			
18.	4	Jenkins Pipeline Script; Jenkins Integration with Dockers	Online Resource [5]
19.	4	Test Stage; Selenium as a Testing Tool to Test Web User Interfaces	Verona: Ch6; Online Resource [6]
20.	4	Setting up a Pipeline in Jenkins having Build and Test Stages over AWS with GitHub and Selenium Integration	Verona: Ch6; Online Resource [5]
21.	5	Introduction to Kubernetes; Kubernetes for Automating Deployment, Scaling, and Management of Containerized Applications	Online Resource [7]
Quiz 03, Assignment 03			
22.	5	Kubernetes Architecture; Master and Worker Nodes; Components of Master and Worker Nodes; Pods; Deployments; Services;	Online Resource [7]

23.	5	Introduction to <i>minikube</i> - A Lightweight Kubernetes Implementation; Introduction to <i>kubectl</i> -A Command-line Tool to Interact with Kubernetes Clusters; <i>minikube</i> Commands; <i>kubectl</i> Commands	Online Resource [8]
24.	5	Infrastructure as Code (IAC); Introduction to YAML Files in Kubernetes;	Online Resource [7]
25.	5	Writing Deployment and Service Kind of YAML Files for Auto Load Balancing	Online Resource [7]
26.	5	Kubernetes Volume Management; Persistent Volume Claim	Online Resource [7]
27.	5	Autoscaling the Application over Kubernetes Cluster	Online Resource [7]
28.	5	Deployment of a Web Application with Auto Load Balancing and Autoscaling Features over <i>minikube</i> Cluster	Online Resource [8]
Quiz 04, Assignment 04			
29.	5	Introduction to Monitoring Stage in DevOps; Issues Tracking	Verona: Ch8, Ch9
30.	5	Prometheus as the Monitoring Tool; Grafana as Visualization Tool	Online Resource [9, 10]
Final Term Exam			

Text and Reference Books

Textbooks:

1. Cloud Computing with Security: Concepts and Practices, Naresh Kumar Sehgal, Pramod Chandra P. Bhatt, John M. Acken, Springer, 2020.
2. Practical DevOps: Implement DevOps in your organization by effectively building, deploying, testing, and monitoring code, Verona, J, Packt Publishing Ltd, 2018.

Reference Books:

1. Cloud computing: implementation, management, and security, Ransome, J. F, CRC press, 2017.
2. Design and Implementation of Service Management in DevOps Enabled Cloud Computing Models. In Handbook of Research on End-to-End Cloud Computing Architecture Design, Eigenbrode, S., & Nassar, S, IGI Global, 2017.
3. Hands-On Serverless Computing, Kuldeep Chowhan, Packt Publishing, 2019.

Online Resources:

- [1] User Management. <https://ubuntu.com/server/docs/security-users>
- [2] Amazon Elastic Compute Cloud Documentation. <https://docs.aws.amazon.com/ec2/>
- [3] Docker Docs. <https://docs.docker.com/>
- [4] Git Documentation. <https://git-scm.com/doc>
- [5] Jenkins User Documentation. <https://www.jenkins.io/doc/>
- [6] The Selenium Browser Automation Project. <https://www.selenium.dev/documentation/>
- [7] Kubernetes Documentation. <https://kubernetes.io/docs/home/>
- [8] minikube start. <https://minikube.sigs.k8s.io/docs/start/>
- [9] Prometheus - Monitoring system & time series database. <https://prometheus.io/>
- [10] Grafana - https://prometheus.io/docs/tutorials/visualizing_metrics_using_grafana/